Check for updates

Geophysical Research Letters

RESEARCH LETTER
10.1029/2024GL110286

Three-Stage India-Asia Collision Proposed by the Thrice Remagnetizations of the Tethyan Himalaya Terrane

Yabo Tong[1,2,3], Junling Pei[4], Tao Qian[1,2], Shengsi Sun[5], Lifu Hou[1], Xinxin Sun[6], Zijian Zhang[6], and Bin Yang[6]

[1]Institute of Geomechanics, Chinese Academy of Geological Sciences, Beijing, China, [2]Key Laboratory of Paleomagnetism and Tectonic Reconstruction, Ministry of Land and Resources, Beijing, China, [3]SinoProbe Laboratory, Chinese Academy of Geological Sciences, Beijing, China, [4]State Key Laboratory of Nuclear Resources and Environment, East China University of Technology, Nanchang, China, [5]State Key Laboratory of Continental Dynamics, Department of Geology, Northwest University, Xi'an, China, [6]College of Resources, Environment and Tourism, Capital Normal University, Beijing, China

Key Points:

- The east-central Tethyan Himalaya Terrane (THT) experienced three phases of large-scale remagnetization during 62.3–50.0 Ma
- The collision of THT with the Lhasa Terrane commenced at 62.3–60.9 Ma and finished at 59.8–58.5 Ma
- India finally collided with the THT at the paleolatitude of ∼10.9 ± 5.1°N at ∼50.0 Ma

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

Y. Tong,
tyb816@163.com

Citation:

Tong, Y., Pei, J., Qian, T., Sun, S., Hou, L., Sun, X., et al. (2024). Three-stage India-Asia collision proposed by the thrice remagnetizations of the Tethyan Himalaya Terrane. *Geophysical Research Letters*, *51*, e2024GL110286. https://doi.org/10.1029/2024GL110286

Received 13 MAY 2024
Accepted 9 JUL 2024

Author Contributions:

Data curation: Yabo Tong, Lifu Hou
Formal analysis: Yabo Tong, Tao Qian, Shengsi Sun
Funding acquisition: Yabo Tong, Junling Pei
Investigation: Yabo Tong, Junling Pei, Lifu Hou, Xinxin Sun, Zijian Zhang, Bin Yang
Methodology: Yabo Tong, Junling Pei, Shengsi Sun
Resources: Yabo Tong
Software: Shengsi Sun
Supervision: Yabo Tong
Validation: Yabo Tong
Visualization: Yabo Tong
Writing – original draft: Yabo Tong

© 2024 The Author(s).
This is an open access article under the terms of the Creative Commons Attribution-NonCommercial-NoDerivs License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

Abstract Crustal deformation and hydrothermal percolation related to the India-Asia collision have caused extensive remagnetization of the Tethyan Himalaya Terrane (THT). The present work identified three phases of regional remagnetization during 62.3–50.0 Ma for the east-central THT. Consequently, a model of three-stage India-Asia collision was proposed. The east-central THT first collided with the southward migrated southern margin of the Lhasa Terrane (LT) at 5.4 ± 0.9°N during 62.3–60.9 Ma. Subsequently, the THT continuously moved northward and pushed the southern margin of the LT back to its original position prior to the initiation of fore-arc and back-arc extension on both sides of the Gangdese magmatic arc. Since the final suturing of the THT with Asia at ∼10°N during 59.8–58.0 Ma, the east-central THT remained stationary until India collided with it at 10.9 ± 5.1°N at ∼50.0 Ma.

Plain Language Summary The collision of India and Asia caused intense tectonic deformation and hydrothermal alteration throughout the Tethyan Himalaya Terrane (THT), which resulted in the large-scale remagnetization in the THT. The regional remagnetization of the THT can be used to constrain the India-Asia collision process, on the premise that the time of remagnetization can be determined. Based on this assumption, we measured two representative Paleocene remagnetized components from Early Jurassic limestones in the Gyangze Basin in the east-central THT. These remagnetized components, combined with non-remagnetized components and remagnetization events recorded in the adjacent areas, suggest that the east-central THT experienced three phases of regional remagnetization during 62.3–50.0 Ma. The first and second phases of remagnetization in the north-central part of the east-central THT occurred at the paleolatitude of 5.4 ± 0.9°N at 62.3–60.9 Ma and 10.3 ± 1.0°N–9.5 ± 1.1°N at 59.8–58.0 Ma, respectively. The third phase of remagnetization occurred in the southern part of the east-central THT, at the paleolatitude of 10.9 ± 5.1°N at ∼50.0 Ma. Consequently, a model of three-stage India-Asia collision and southward spreading tectonic deformation of the THT was proposed based on these successive remagnetizations.

1. Introduction

The collision of India and Asia in the Paleocene–Eocene contributed to the uplift of the Tibetan Plateau (Ding et al., 2022; Tapponnier et al., 2001), and significantly impacted global climate change (Kent & Muttoni, 2008; Raymo & Ruddiman, 1992; Wu et al., 2022). In the last three decades, geological and paleomagnetic studies have frequently been carried out in the Lhasa Terrane (LT) and the Tethyan Himalaya Terrane (THT) to constrain the India-Asia collision process (e.g., Patzelt et al., 1996; Yin & Harrison, 2000).

In traditional paleomagnetic studies, the initial collision between India and Asia was constrained by the overlapping of paleolatitudes between the northern margin of India and the southern margin of Asia. However, the India-Asia collision model remains hotly debated, with different interpretations being proposed. The traditional single collision model suggested that Greater India, with an N-S extent of 2000–3,000 km and Asia (Meng et al., 2020, 2023) or ∼900 km (Jin et al., 2023; Yang et al., 2019) directly collided with Asia. The two-stage collision model suggested a first collision between the THT and the LT (e.g., van Hinsbergen et al., 2012; Yang et al., 2015; Yuan et al., 2021) or between an island arc system and India (Aitchison et al., 2007), or between an island arc system and LT (Pusok & Stegman, 2020), followed by the final India-Asia collision. The three-stage

Writing – review & editing: Yabo Tong, Junling Pei, Tao Qian

collision model even suggested a first collision between the THT and island arc, followed by the successive collision of the THT-island arc with Asia and India with THT (Jadoon et al., 2021; Yuan et al., 2022). Consequently, the time of India-Asia collision fell within a wide age range of ∼65.0–47.0 Ma, and the post-collision latitudinal crustal convergence in Asia changed from ∼1,700 km to ∼200 km (e.g., Huang et al., 2013; Ma et al., 2017; Sun et al., 2012; Tan et al., 2010; Tong et al., 2022; van Hinsbergen et al., 2012; Yang et al., 2015). One of the primary reasons for these controversies is the different interpretations of the paleolatitude and morphological characteristics of the northern margin of India during the Late Cretaceous and Paleocene. However, the current paleomagnetic studies within the THT cannot resolve these disputes due to the potential remagnetization and disordered stratigraphic sequence of studied strata in the THT (Huang et al., 2015; Tong et al., 2019).

The India-Asia collision led to intense tectonic deformation and hydrothermal activity throughout the THT, resulting in large-scale remagnetization of strata in the THT (Appel et al., 2012; Besse et al., 1984; Huang et al., 2015, 2017a, 2017b; Klootwijk & Bingham, 1980; Tong et al., 2008; Xu et al., 2022). The stage and timing of the India-Asia collision during the entire suturing process could be generally consistent with that of the regional remagnetization in the THT. Thus, the spatio-temporal process of the India-Asia collision could be constrained by dating the acquisition times of remagnetized components from the Jurassic to Paleocene strata of the THT. Based on this concept, we conducted a paleomagnetic study on Early Jurassic limestones, which are known to be susceptible to remagnetization. Here, we obtained two representative Paleocene remagnetized components from the Gyangze Basin in the east-central THT. These results, integrated with interpreted primary magnetic components and remagnetization events recorded by Upper Cretaceous and Paleogene strata in the adjacent areas, enable us to constrain the entire suturing process of India-Asia.

2. Geology and Sampling

The Gyangze Basin is situated on the northern margin of the east-central THT (Figure 1a). Mesozoic and Paleocene sedimentary strata are widespread in this basin. The Paleocene strata are termed the Jiachala Formation; the Cretaceous strata are divided into the Lower Cretaceous Jiabula Formation and the Upper Cretaceous Zongzhuo Formation; and the Jurassic strata are divided into the Lower Jurassic Ridang, the Middle Jurassic Zhela, and the Upper Jurassic Weimei Formations (Bureau of Geology and Mineral Resources of Xizang Province, 1993; Wu et al., 2014) (Figure 1b).

The Upper Jurassic Weimei Formation consists mainly of quartz sandstone and sandy shale, which conformably overlies the dark-gray and gray sandy shale, siltstone, and feldspathic quartz sandstone of the Middle Jurassic Zhela Formation. The Zhela Formation conformably overlies the Lower Jurassic Ridang Formation. The Ridang Formation mainly consists of gray-black shale, calcareous shale, and argillaceous limestone. Ammonoid fossils occur widely in the Ridang Formation. They are subdivided into three assemblages from bottom to top: the *Psiloceras-Schlothemia* assemblage in the lower segment; the *Arietites-Sulciferites* assemblage in the middle segment; and the *Prodactylioceras-Galaticeras* assemblage in the upper segment. The three ammonoid fossil assemblages consistently indicate an Early Jurassic age for the Ridang Formation in the Gyangze area (Bureau of Geology and Mineral Resources of Xizang Province, 1993). The detrital zircon studies in the Middle Jurassic Zhela and Upper Jurassic Weimei Formations sandstone in the Gyangze area constrained the Middle and Late Jurassic ages for the two Formations and revealed the affinity between the THT and India (Jin et al., 2024). These results further confirmed the Early Jurassic age for the Ridang Formation. A total of 21 sites, comprising ∼220 core samples of argillaceous limestone of the Lower Jurassic Ridang Formation, were collected from a section in the Gyangze Basin (Figure 1b).

3. Rock Magnetic, Petrographic, and Paleomagnetic Results

Rock magnetic results associated with scanning electron microscopic observations and energy dispersive spectrometry analyses suggest that single domain (SD)/pseudo single domain (PSD) pyrrhotite and greigite are the predominant magnetic carrier for argillaceous limestone of the Lower Jurassic Ridang Formation (See Methods, and Figures S1 and S2 in Supporting Information S1). In most cases, greigite is associated with pyrite (Figures S2c, S2d, and S2f in Supporting Information S1) or pyrrhotite (Figures S2b and S2e in Supporting Information S1s) within framboid aggregates, suggesting the transformations of pyrrhotite to greigite, and then to pyrite, within the reduction conduction (Berner, 1984; Cutter & Velinsky, 1988; Muramoto et al., 1991).

19448007, 2024, 15, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024GL110286 by Curtin University Library, Wiley Online Library on [13/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

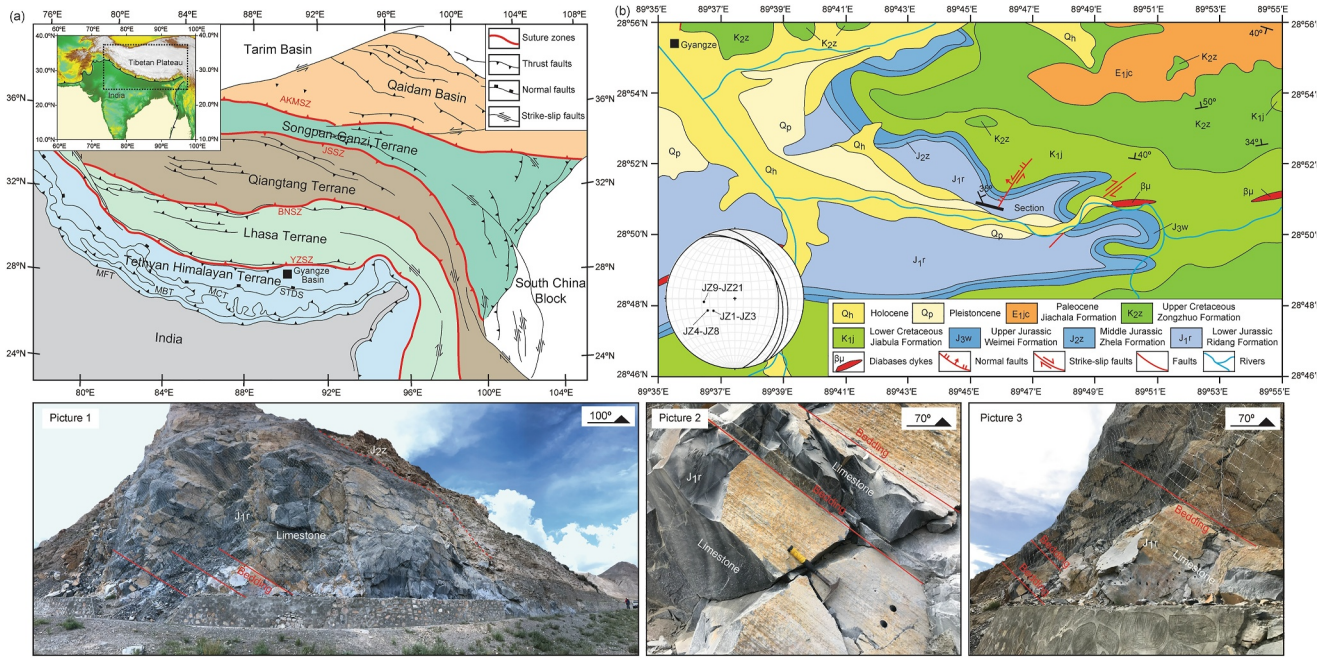


Figure 1. (a) Schematic tectonic geological map of the Tibetan Plateau. (b) Schematic geological map of the Gyangze Basin, showing the location of the paleomagnetic sampling section in the Early Jurassic Ridang Formation. Picture 1 shows the sampling section. Pictures 2 and 3 show the bedding and lithology of the sampled Early Jurassic Ridang Formation. The black square in panel (a) is our sampling area. The block rectangle in panel (b) is our sampling section. AKMSZ, Ayimaqing-Kunlun-Muztagh suture zone; JSSZ, Jinshajiang suture zone; BNSZ, Bangong-Nujiang suture zone; YZSZ, Yarlung-Zangbo suture zone; STDS, South Tibet Detachment System; MCT, Main Central Thrust; MBT, Main Boundary Thrust; MFT, Main Frontal Thrust.

Limestone specimens were subjected to stepwise thermal demagnetization. Remagnetized components A and B were defined by the linear decay of magnetization within the temperature interval from ~200°C to ~400°C, and ~300°C to ~350°C, respectively (see Methods and Figures S3 in Supporting Information S1). The specimen-mean direction of magnetic component A is $D_g = 341.3°$, $I_g = 19.5°$, $k = 76.2$, $\alpha_{95} = 1.9°$, $n = 71$ (specimens) before tilt correction, and $D_s = 351.4°$, $I_s = 13.6°$, $k = 62.2$, $\alpha_{95} = 2.2°$ after tilt correction (Figure S3e in Supporting Information S1). The specimen-mean direction of component B is $D_g = 163.6°$, $I_g = -9.5°$, $k = 67.9$, $\alpha_{95} = 1.9°$, $n = 87$ (specimens) before tilt correction and $D_s = 171.1°$, $I_s = -13.0°$, $k = 65.0$, $\alpha_{95} = 1.9°$ after tilt correction (Figure S3f in Supporting Information S1). The normal polarity magnetic component A is approximately antipodal with the reversed polarity magnetic component B after tilt correction. However, the two components can be isolated from the same specimens of site JZ4, over different temperature intervals (Figure S3b in Supporting Information S1). The coexistence of magnetic components A and B in site JZ4 indicates that they are two independent components. The remagnetized components A and B could be carried by greigite and pyrrhotite, respectively, related to a tectonically driven Sulfur-containing fluid migration event (Huang et al., 2015, 2017a; Weaver et al., 2002).

The sulfate reduction sequentially forms a series of iron sulfides, from pyrrhotite to greigite, and finally to pyrite, in the reduction conduction (Berner, 1984). This suggests that the remagnetized component B carried by pyrrhotite, was acquired earlier than the remagnetized component A, which was carried by greigite. After applying the syn-tilting Fisher analyses for the two remagnetized components, the k_{max} is only slightly larger than that after tilt correction, indicating that the two remagnetized components were acquired during the final stage of the folding. Therefore, the D_{max} and I_{max} of the two remagnetized components after applying the syn-tilting Fisher analyses are used for discussion (Figures S3e and S3f, Table S1 in Supporting Information S1).

4. Timing of Regional Remagnetization of the East-Central THT

The remagnetized components A and B of the Lower Jurassic Ridang Formation from the Gyangze Basin suggest that the north-central part of the east-central THT was remagnetized when it situated at the paleolatitude of 10.3 ± 1.0°N during the late stage of folding and at 5.4 ± 0.9°N during an earlier time (Table S2 in Supporting

19448007, 2024, 15, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024GL110286 by Curtin University Library, Wiley Online Library on [13/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

10.1029/2024GL110286

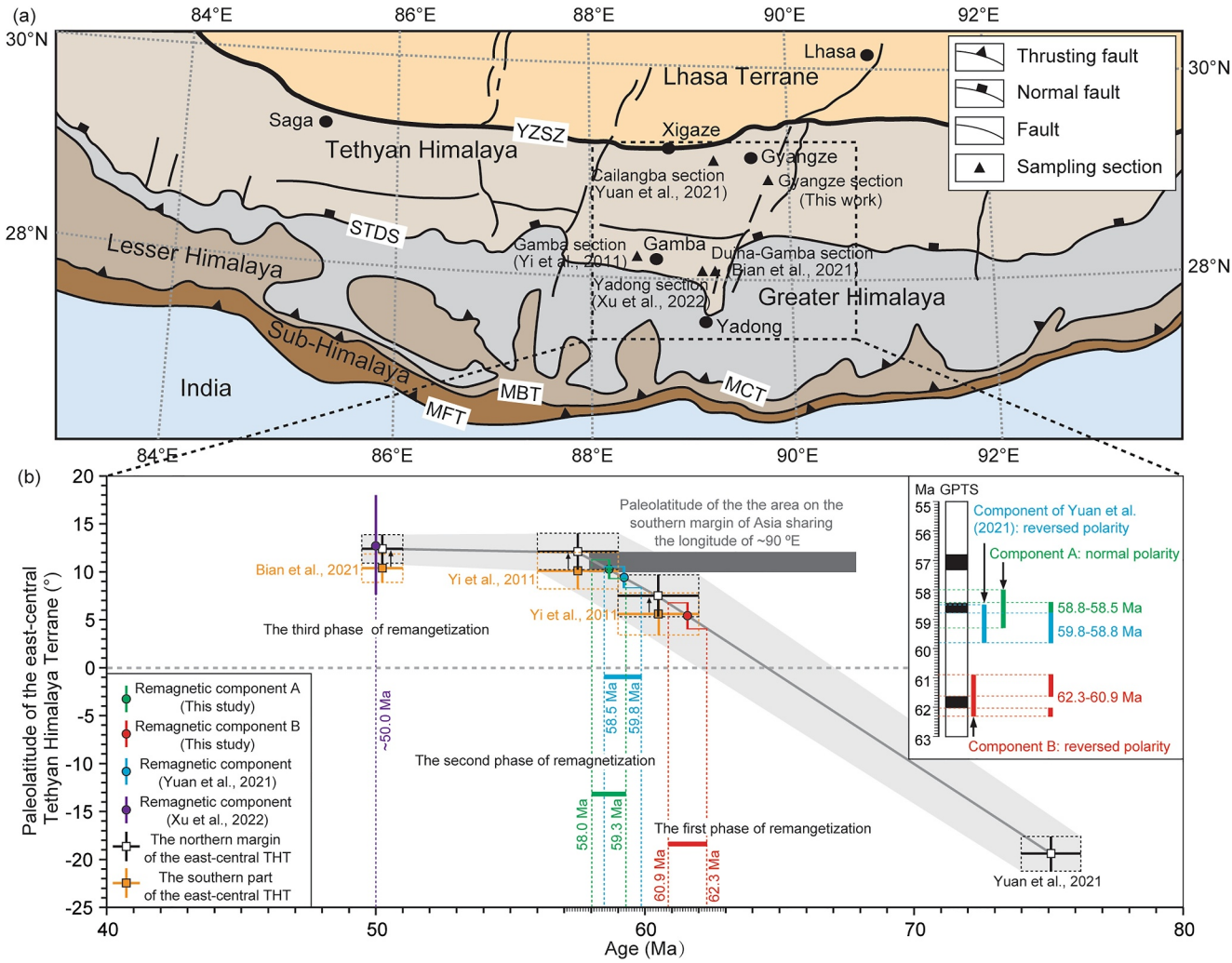


Figure 2. (a) Locations of paleomagnetic studies on Jurassic to Eocene strata in the Tethyan Himalaya Terrane (THT). (b) Variation of paleolatitudes of the north-central part of the east-central THT during ~76.0–50.0 Ma, which was used to constrain the timing of the remagnetizations of the east-central THT. The paleomagnetic data sets were listed in Table S2 in Supporting Information S1.

Information S1). In previous paleomagnetic studies, primary remanence components were obtained from Upper Cretaceous, Paleocene, and Eocene strata in the east-central THT (Figure 2a, Table S2 in Supporting Information S1). The study area of the present work was selected as the reference point (28.8°N/89.7°E) to calculate the paleo-latitudes and escribe the movement process of the of the east-central THT, by using these remagnetized and primary remanence components (Figure 2b). We describe these studies as follows.

The primary remanence component from the red chert dated to 76.2–74.0 Ma in the Cailangba area in the northwestern part of the Gyangze Basin suggests that this basin was situated at 19.4 ± 1.8°S during 76.2–74.0 Ma (Yuan et al., 2021) (Figure 2b). Yi et al. (2011) reported high-quality paleomagnetic results of the Paleocene Zongpu Formation from the Gamba area, yielding paleolatitudes of 5.6 ± 2.2°N at 62.0–59.0 Ma and 10.1 ± 1.9°N at 59.0–56.0 Ma. However, Huang et al. (2017a, 2017b) suggested that most of the Cretaceous and Paleocene limestones of the Dingri and Gamba areas were remagnetized via orogenic fluids. The high-resolution petrographic studies of Zhao et al. (2021) identified a large amount of multi-to single domain detrital magnetite and nanoscale biogenic magnetite in the Paleocene limestones of the Gamba area, suggested as the main magnetic carriers for the primary remanence component of limestones of the Zongpu Formation in the Gamba area. In consideration of the plentiful high-resolution petrographic evidences of Zhao et al. (2021), here we adopt the interpretation of Zhao et al. (2021), and thus the conclusions of Yi et al. (2011) were used for further discussion.

19448007, 2024, 15, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024GL110286 by Curtin University Library, Wiley Online Library on [13/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

19448007, 2024, 15, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024GL110286 by Curtin University Library, Wiley Online Library on [13/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Bian et al. (2021) isolated a primary remanence component from the Zhepure Formation limestone dated to 51.0–49.5 Ma in the Duina-Gamba area, which provides a paleolatitude of 14.6 ± 1.7°N for the east-central THT, after applying an E/I inclination shallowing correction (Tauxe, 2005; Tauxe & Kent, 2004). However, the results from the sampling section of Bian et al. (2021), situated in the two limbs of a plunging fold, show that E/I inclination shallowing correction resulted in an excessive degree of correction. The remanence component carried by detrital magnetite is often a post-depositional detrital remanent magnetization (pDRM) in most limestones, and the pigment in red and pink limestones carries a chemical remanent magnetization that is essentially contemporaneous with the pDRM carried by the detrital magnetite (Butler, 1998). Thus, the limestone has not suffered from depositional and compaction-induced inclination shallowing. For these reasons, we use the non-E/I inclination shallowing corrected results of Bian et al. (2021) for further discussion, which shows that the Duina-Gamba area was situated at 10.4 ± 1.5°N at 51.0–49.5 Ma.

The THT, currently with a width of ∼120 km from south to north, was shortening by ∼ 85–∼139 km during the Cenozoic, based on the shortening estimations for the Tethyan Himalayan sequence on the north side of the South Tibetan detachment system (Parsons et al., 2020), indicating that the northern margin of the east-central THT should be located at ∼205–259 km (∼2°) north of the southern margin of the east-central THT. Therefore, the northern margin of the east-central THT was constrained to ∼7.6 ± 2.2°N at 62.0–59.0 Ma, ∼12.1 ± 1.9°N at 59.0–56.0 Ma and ∼12.4 ± 1.5°N at 51.0–49.5 Ma (Figure 2b), based on the paleomagnetic results from the southern margin of the east-central THT (Bian et al., 2021; Yi et al., 2011).

These paleomagnetic data sets suggest that the northern margin of the east-central THT unidirectionally moved northward, from 19.4 ± 1.8°S during 76.2–74.0 Ma to 7.6 ± 2.2°N during 62.0–59.0 Ma, and then to 12.1 ± 1.9°N during 59.0–56.0 Ma. It then remained stationary at ∼12.4 ± 1.5°N, until at least 51.0–49.5 Ma. These results confirmed that the THT experienced northward unidirectional movement with a stabilized average rate during 76.2–56.0 Ma, which permits us to constrain the timing of remagnetization of the THT based on comparisons of the paleolatitudes calculated from the remagnetized components and primary magnetic components (Figure 2b).

The paleolatitude of 5.4 ± 0.9°N of the Gyangze area constrained by the remagnetized component B is slightly lower than the paleolatitude of 7.6 ± 2.2°N during 62.0–59.0 Ma, suggesting that the remagnetized component B was possibly acquired during 62.3–60.9 Ma (Figure 2b). The paleolatitude of 10.3 ± 1.0°N of the Gyangze area constrained by the remagnetized component A is approximately consistent with that of 9.5 ± 1.1°N calculated from the remagnetized component of Yuan et al. (2021), obtained from limestone dated to 76.2–74.0 Ma in the Cailangba area. These paleolatitudes are higher than the paleolatitude of 7.6 ± 2.2°N during 62.0–59.0 Ma, but they are slightly lower than 12.1 ± 1.9°N during 59.0–56.0 Ma, suggesting that they were acquired during the period of 59.8–58.0 Ma (Figure 2b). Considering that the remagnetized component B and the remagnetized component of Yuan et al. (2021) are of reversed polarity, and that the remagnetized component A is of normal polarity, their acquisition times can be further corrected based on a comparison with the geomagnetic polarity time scale (Gradstein et al., 2020) (Figure 2b). The comparisons show that the remagnetized components A and B were acquired during 58.8–58.5 Ma and 62.3–60.9 Ma, respectively, and that the remagnetized component of Yuan et al. (2021) was acquired during 59.8–58.8 Ma. The pre-folding remagnetized component dated to ∼50.0 Ma from the Yadong area provided a paleolatitude of 10.9 ± 5.1°N for the southern part of the east-central THT (Xu et al., 2022), suggesting that the northern margin of the east-central THT was situated at ∼12.9 ± 5.1°N at ∼50.0 Ma.

Based on these analyses, we proposed that the north-central part of the east-central THT experienced the first phase of remagnetization at 5.4 ± 0.9°N at 62.3–60.9 Ma, followed by the second phase of remagnetization at 10.3 ± 1.0°N–9.5 ± 1.1°N during 59.8–58.0 Ma. In addition, the third phase of remagnetization occurred at the southern part of the east-central THT, at ∼10.9 ± 5.1°N at ∼50.0 Ma (Figure 2b). However, we need to admit that the attempt to estimate the paleolatitude-time distribution of remagnetized limestones could give speculative results, because of the system errors of paleomagnetic data sets from the east-central THT. Thus, it is essential to obtain more reliable Cretaceous and Paleogene paleomagnetic data sets from the THT in the future, in order to establish the precise Apparent Polar Wander Paths of the THT.

A later stage of remagnetization at ∼22 Ma was observed in the Cretaceous dykes in the Nagarze area near the Gyangze Basin (Antolin et al., 2010). They suggested that pyrrhotite is the magnetic carrier for the post-folding remagnetized component of the east-central THT, which is likely of thermoremanent or thermos-chemical origin related to the activity of major faults like Main Central Thrust (MCT) and South Tibetan Detachment in the THT,

since the Early Miocene. However, the collision of India-Asia can also induce the quasi-synchronous crustal deformation on both sides of the suture zone between India and Asia, resulting in the early stage remagnetization in the east-central THT. Thus, we proposed that the THT experienced the early stage remagnetization related to the collision of India-Asia and the later stage remagnetization related to the activity of major faults in the THT since the Early Miocene.

5. Three-Stage of India-Asia Collision

The paleomagnetic data from the northern side of the Gangdese magmatic arc suggest that the southern LT remained stationary and did not experience latitudinal migration during the Late Cretaceous (Ma et al., 2017, 2018; Tong et al., 2019, 2022; Yi et al., 2015). However, the Gangdese magmatic arc on the southern margin of the LT experienced both fore-arc and back-arc extension during the Late Cretaceous and Early Paleocene, forming the Xigaze forearc basin and even opened oceanic basins on both sides of the Gangdese magmatic arc (Kapp & DeCelles, 2019; Wang et al., 2017), indicating that the southern margin of the LT has experienced substantial southward migration prior to the India-Asia collision. The difference in paleomagnetic data sets from the Paleocene Sangdanlin Formation in the Saga area, on the northern margin of the central THT argued that whether the THT rifted from northern India, forming the North India Sea (Meng et al., 2023; Yuan et al., 2021), which further induced the debate of two-stage or single-stage India-Asia collision. The recent nature and density structure constraining with numerical models for the India and Tibet proposed that the northern margin of India contained a ∼1,300 km long thinned continental margin covered by seawater, and a ∼600 km long external buoyant continental terrane (Liu et al., 2023). This model proposed that although northern India experienced significant rifting, resulting in the separation of the THT from the northern India, it formed a ∼1,300 km long thinned continental margin between the THT and the Indian Craton, rather than opening the North India Sea. This assumption reconciled the divergences of paleomagnetic studies, sedimentary data, and the structural geology along the suture zone between India and Asia.

The north-central part of the east-central THT experienced the first phase of remagnetization at 5.4 ± 0.9°N, suggesting that the THT collided with the southward migrated southern margin of the LT at this paleolatitude during 62.3–60.9 Ma (Figure 3a), leading to the folding deformation of the north-central part of the THT (Figure 3b). The reconstructions of the paleolatitude and morphological characteristics for the LT suggested that the southern LT linearly extended in the direction of ∼317° before the India-Asia collision (Tong et al., 2019, 2022; Yi et al., 2015). This reconstruction suggests that the southern margin of the LT shared the same longitude of ∼90°E with the Gyangze Basin and the Duina-Gamba Basin was situated at a paleolatitude of ∼10.0–12.0°N during the Late Cretaceous and Paleocene. This paleolatitude overlapped with the observed paleolatitude of 10.3 ± 1.0°N–9.5 ± 1.1°N during 59.8–58.5 Ma for the second phase of remagnetization along the north-central part of the east-central THT. Thus, after suturing of the THT with the migrated southern margin of the LT, the THT continuously moved northwards, which terminated the fore-arc and back-arc extension on both sides of the Gangdese magmatic arc, closed the Xigaze forearc basin and pushed the southern margin of Asia back to its original paleolatitude (∼10.0–12.0°N) prior to the southward migration, indicating that the final collision between the THT and Asia had occurred at 59.8–58.5 Ma (Figure 3a). This process further intensified the folding deformation of the north-central part of the THT (Figure 3b). During 51.0–49.5 Ma, the northern margin of the east-central THT was at ∼12.4 ± 1.5°N, which is the same as the southern margin of the LT during 59.8–58.5 Ma, indicating that the northern margin of the THT approximately remained stationary since its collision with Asia at 59.8–58.5 Ma (Figure 3a). This probably suggests that the collision of the THT with Asia had already finished prior to 51.0–49.5 Ma.

During the entire collision process of the THT with the LT, the crustal folding deformation and remagnetization were restricted in the northern margin of the THT (Figure 3b). The third phase of remagnetization occurred in the southern part of the east-central THT, at the paleolatitude of 10.9 ± 5.1°N at ∼50.0 Ma (Xu et al., 2022), which is consistent with the stabilized paleolatitude of the southern part of the east-central THT during 59.8–58.5 Ma to 51.0–49.5 Ma. This suggested that the suturing of the Indian Carton with the stabilized THT initiated intense crustal folding, subduction and delaminating of the ∼1,300 km long thinned continental margin between the main body of India and the THT, and the widespread remagnetization in the southern part of the east-central THT at ∼50.0 Ma (Figures 3a and 3b).

19448007, 2024, 15, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024GL110286 by Curtin University Library, Wiley Online Library on [13/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

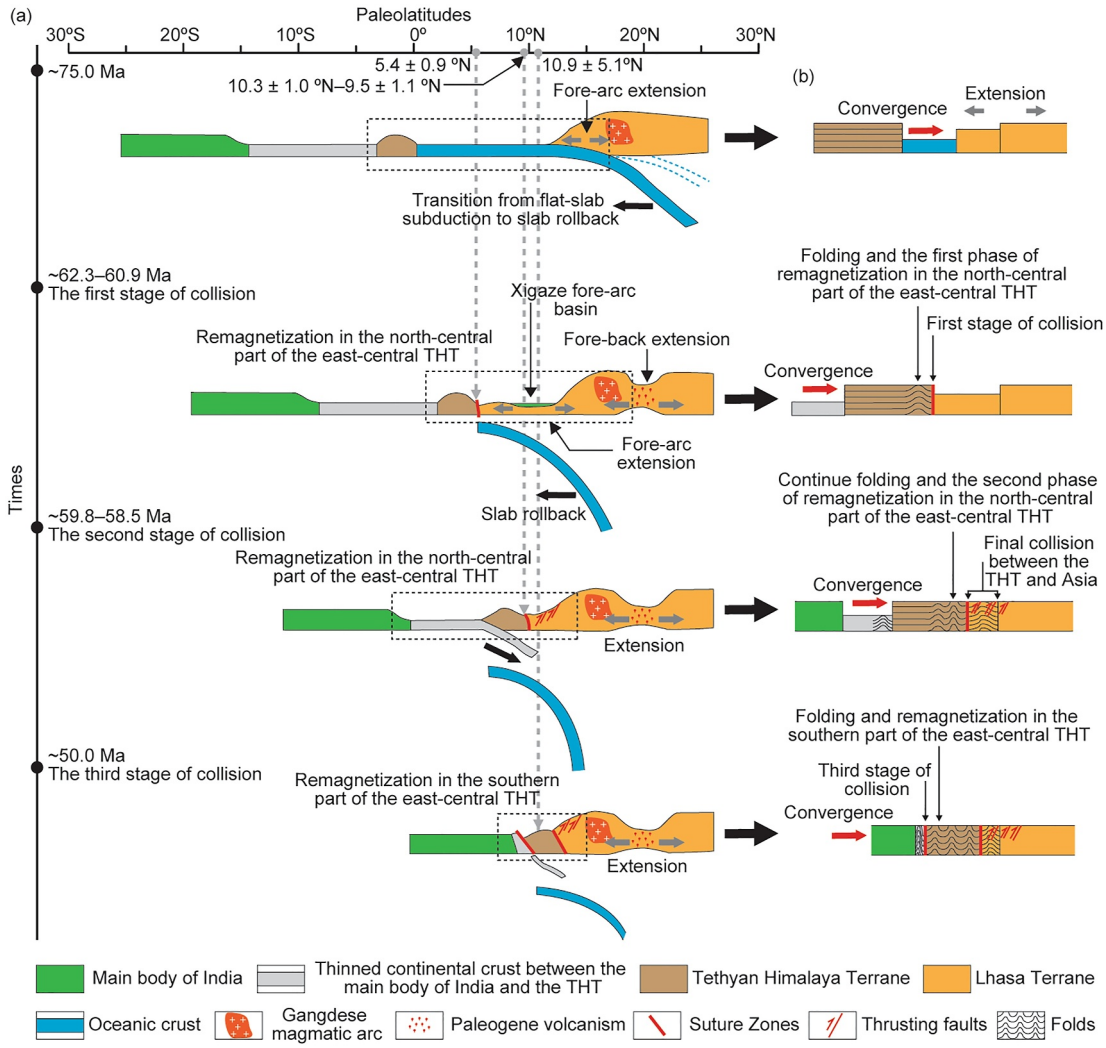


Figure 3. (a) The three-stage India-Asia collision, constrained by the analysis of the original magnetic and remagnetized components from the Jurassic and Eocene strata in the east-central Tethyan Himalaya Terrane (THT) and the Lhasa Terrane. (b) The process of tectonic deformation and widespread remagnetization in the east-central THT related to the India-Asia collision.

6. Conclusion

The east-central THT experienced three phases of regional remagnetization during 62.3–50.0 Ma. The first and second phases of remagnetization occurred in the north-central part of the east-central THT, at the paleolatitude of 5.4 ± 0.9°N at 62.3–60.9 Ma and 10.3 ± 1.0°N–9.5 ± 1.1°N at 59.8–58.0 Ma, respectively. The third phase of remagnetization occurred in the southern part of the east-central THT, at the paleolatitude of 10.9 ± 5.1°N at ~50.0 Ma. A model of three-stage India-Asia collision was proposed by the three phase remagnetizations. The THT first collided with the southward migrated southern margin of the LT at 5.4 ± 0.9°N at 62.3–60.9 Ma. Subsequently, the THT continuously moved northward and pushed the southern margin of Asia back to its original position prior to the initiation of fore-arc and back-arc extension on both sides of the Gangdese magmatic arc, followed by the second collision between the THT and Asia at 10.3 ± 1.0°N–9.5 ± 1.1°N during 59.8–58.0 Ma. The THT has finished suturing with the LT and remained stationary during 59.8–49.5 Ma. Finally, the main body of India collided with the THT at ~50.0 Ma, which synchronously induced the tectonic deformation in the southern part of the east-central THT. The three-stage India-Asia collision of the present study suggests that the THT did not first collide with an island arc, but it sutured with the southward migrated southern margin of Asia, which is the most significant difference with previous three-stage India-Asia collision (Jadoon et al., 2021; Yuan et al., 2022).

19448007, 2024, 15, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024GL110286 by Curtin University Library, Wiley Online Library on [13/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Data Availability Statement

Data for thermal demagnetization and rock magnetic results for the Jurassic limestone of the Gyangze Basin is available at Tong and Pei (2024).

Acknowledgments
We thank professors Baochun Huang and Peter C. Lippert for their insightful discussions and suggestions, which have improved this work. Dr. Yiming Ma, professor Erwin Appel and another anonymous reviewer are appreciated for reviewing this manuscript. This work was supported by the Second Tibetan Plateau Scientific Expedition and Research Program (2019QZKK0901) and the National Natural Science Foundation of China (91855216).

References

Aitchison, J. C., Ali, J. R., & Davis, A. M. (2007). When and where did India and Asia collide? *Journal of Geophysical Research*, *112*(B5), B05432. https://doi.org/10.1029/2006jb004706

Antolin, B., Appel, E., Gloaguen, R., Dunkl, I., Ding, L., Montomoli, C., et al. (2010). Paleomagnetic evidence for clockwise rotation and tilting in the eastern Tethyan Himalaya (SE Tibet): Implications for the Miocene tectonic evolution of the NE Himalaya. *Tectonophysics*, *493*(1–2), 172–186. https://doi.org/10.1016/j.tecto.2010.07.015

Appel, E., Crouzet, C., & Schill, E. (2012). Pyrrhotite remagnetizations in the Himalaya: A review. *Geological Society London Special Publications*, *371*(1), 163–180. https://doi.org/10.1144/sp371.1

Berner, B. A. (1984). Sedimentary pyrite formation: An update. *Geochimica et Cosmochimica Acta*, *48*(4), 605–615. https://doi.org/10.1016/0016-7037(84)90089-9

Besse, J., Courtillot, V., Pozzi, J. P., Westphal, M., & Zhou, Y. X. (1984). Palaeomagnetic estimates of crustal shortening in the Himalayan thrusts and Zangbo suture. *Nature*, *311*(5987), 621–626. https://doi.org/10.1038/311621a0

Bian, W. W., Yang, T. S., Peng, W. X., Wang, S., Gao, F., Zhang, S. H., et al. (2021). Paleomagnetic constraints on the India-Asia collision and the size of greater India. *Journal of Geophysical Research: Solid Earth*, *126*(6), e2021JB021965. https://doi.org/10.1029/2021JB021965

Bureau of Geology and Mineral Resources of Xizang Province (BGMRX). (1993). *Regional geology of Xizang (Tibet) autonomous region* (pp. 166–170). Geological Publishing House. (in Chinese).

Butler, R. F. (1998). *Paleomagnetism: Magnetic domains to geologic Terranes* (electronic-ed., pp. 146–147). A.Z., Department of Geosciences.

Cutter, G. A., & Velinsky, D. J. (1988). Temporal variations of sedimentary sulfur in a Delaware salt marsh. *Marine Chemistry*, *23*(3–4), 311–327. https://doi.org/10.1016/0304-4203(88)90101-6

Ding, L., Kapp, P., Cai, F. L., Garzione, C. N., Xiong, Z. Y., Wang, H. Q., & Wang, C. (2022). Timing and mechanisms of Tibetan Plateau uplift. *Nature Reviews Earth & Environment*, *3*(10), 652–667. https://doi.org/10.1038/s43017-022-00318-4

Gradstein, F. M., Ogg, J. G., Schmitz, M. D., & Ogg, G. M. (2020). *The geologic time scale 2020* (pp. 1141–1215). Elsevier. https://doi.org/10.1016/B978-0-12-824360-2.00029-2

Huang, W., Lippert, P. C., Jackson, M. J., Dekkers, M. J., Zhang, Y., Li, J., et al. (2017a). Remagnetization of the Paleogene Tibetan Himalayan carbonate rocks in the Gamba area: Implications for reconstructing the lower plate in the India-Asia collision. *Journal of Geophysical Research: Solid Earth*, *122*(2), 808–825. https://doi.org/10.1002/2016jb013662

Huang, W. T., Dupont-Nivet, G., Lippert, P. C., van Hinsbergen, D. J. J., & Hallot, E. (2013). Inclination shallowing in Eocene Linzizong sedimentary rocks from Southern Tibet: correction, possible causes and implications for reconstructing the India–Asia collision. *Geophysical Journal International*, *194*(3), 1390–1411. https://doi.org/10.1093/gji/ggt188

Huang, W. T., Lippert, P. C., Zhang, Y., Jackson, M. J., Dekkers, M. J., Hu, X. M., et al. (2017b). Remagnetization of carbonate rocks in southern Tibet: Perspectives from rock magnetic and petrographic investigations. *Journal of Geophysical Research: Solid Earth*, *122*(4), 2434–2456. https://doi.org/10.1002/2017jb013987

Huang, W. T., van Hinsbergen, D. J. J., Dekkers, M. J., Garzanti, E., Dupont-Nivet, G., Lippert, P. C., et al. (2015). Paleolatitudes of the Tibetan Himalaya from primary and secondary magnetizations of Jurassic to Lower Cretaceous sedimentary rocks. *Geochemistry, Geophysics, Geosystems*, *16*(1), 77–100. https://doi.org/10.1002/2014GC005624

Jadoon, U. F., Huang, B. C., Shah, S. A., Rahim, Y., Khan, A. A., & Bibi, A. (2021). Multi-stage India-Asia collision: Paleomagnetic constraints from Hazara-Kashmir syntaxis in the western Himalaya. *Geological Society of America Bulletin*, *134*(5–6), 1109–1128. https://doi.org/10.1130/B36116.1

Jin, S. C., Sun, X. X., Jing, X. Q., Zhang, Z. J., Zhang, X., & Yang, Z. Y. (2023). A smaller greater India and a middle-early Eocene collision with Asia. *Geophysical Research Letters*, *50*(3), e2022GL101372. https://doi.org/10.1029/2022GL101372

Jin, S. C., Tong, Y. B., Sun, X. X., Zhang, Z. J., Pei, J. L., Hou, L. F., & Yang, Z. Y. (2024). Extensional tectonics of the Indian passive continental margin in the Middle and Late Jurassic: Constraints from detrital zircon ages in the eastern Tethyan Himalaya. *Journal of Geodynamics*, *159*, 102019. https://doi.org/10.1016/j.jog.2024.102019

Kapp, P., & DeCelles, P. (2019). Mesozoic-Cenozoic geological evolution of the Himalayan-Tibetan orogen and working tectonic hypotheses. *American Journal of Science*, *319*(3), 159–254. https://doi.org/10.2475/03.2019.01

Kent, D. V., & Muttoni, G. (2008). Equatorial convergence of India and early Cenozoic climate trends. *Proceedings of the National Academy of Sciences of the United States of America*, *105*(42), 16065–16070. https://doi.org/10.1073/pnas.0805382105

Klootwijk, C. T., & Bingham, D. K. (1980). The extent of greater India, III. Palaeomagnetic data from the Tibetan sedimentary series, Thakkhola region, Nepal Himalaya. *Earth and Planetary Science Letters*, *51*(2), 381–405. https://doi.org/10.1016/0012-821X(80)90219-8

Liu, L., Liu, L. J., Morgan, J. P., Xu, Y. G., & Cheng, L. (2023). New constraints on Cenozoic subduction between India and Tibet. *Nature Communications*, *14*(1), 1963. https://doi.org/10.1038/s41467-023-37615-5

Ma, Y. M., Yang, T. S., Bian, W. W., Jin, J. J., Wang, Q., Zhang, S. H., et al. (2018). A stable southern margin of Asia during the Cretaceous: Paleomagnetic constraints on the Lhasa-Qiangtang collision and maximum width of the Neo-Tethys. *Tectonics*, *37*(10), 3853–3876. https://doi.org/10.1029/2018TC005143

Ma, Y. M., Yang, T. S., Bian, W. W., Jin, J. J., Wang, Q., Zhang, S. H., et al. (2017). Paleomagnetic and geochronologic results of latest Cretaceous lava flows from the Lhasa terrane and their tectonic implications. *Journal of Geophysical Research: Solid Earth*, *122*(11), 8786–8809. https://doi.org/10.1002/2017JB014743

Meng, J., Gilder, S. A., Li, Y., Wang, C., & Liu, T. (2020). Expanse of Greater India in the late Cretaceous. *Earth and Planetary Science Letters*, *542*, 116330. https://doi.org/10.1016/j.epsl.2020.116330

Meng, J., Gilder, S. A., Tan, X. D., Li, X., Li, Y. A., Luo, H., et al. (2023). Strengthening the argument for a large Greater India. *PNAS*, *120*(33), e2305928120. https://doi.org/10.1073/pnas.2305928120

Muramoto, J., Honjo, S., Fry, B., Hay, B. J., Howarth, R. W., & Cisne, J. L. (1991). Sulfur, iron and organic carbon fluxes in the Black Sea: Sulfur isotopic evidence for origin of sulfur fluxes. *Deep Sea Research Part A. Oceanography Research Papers*, *38*, S1151–S1187. https://doi.org/10.1016/S0198-0149(10)80029-9

19448007, 2024, 15, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024GL110286 by Curtin University Library, Wiley Online Library on [13/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Parsons, A. J., Hosseini, K., Palin, R. M., & Sigloch, L. (2020). Geological, geophysical and plate kinematic constraints for models of the India-Asia collision and the post-Triassic central Tethys oceans. *Earth-Science Reviews*, *208*, 103084. https://doi.org/10.1016/j.earscirev.2020.103084

Patzelt, A., Li, H. M., Wang, J. D., & Appel, E. (1996). Palaeomagnetism of Cretaceous to Tertiary sediments from southern Tibet: Evidence for the extent of the northern margin of India prior to the collision with Eurasia. *Tectonophysics*, *259*(4), 259–284. https://doi.org/10.1016/0040-1951(95)00181-6

Pusok, A. E., & Stegman, D. R. (2020). The convergence history of India-Eurasia records multiple subduction dynamic processes. *Science Advances*, *6*(19), eaaz8681. https://doi.org/10.1126/sciadv.aaz8681

Raymo, M. E., & Ruddiman, W. F. (1992). Tectonic forcing of late Cenozoic climate. *Nature*, *359*(6391), 117–122. https://doi.org/10.1038/359117a0

Sun, Z. M., Pei, J. L., Li, H. B., Xu, W., Jiang, W., Zhu, Z. M., et al. (2012). Paleomagnetism of late Cretaceous sediments from southern Tibet: Evidence for the consistent paleolatitudes of the southern margin of Eurasia prior to the collision with India. *Gondwana Research*, *21*(1), 53–63. https://doi.org/10.1016/j.gr.2011.08.003

Tan, X. D., Gilder, S., Kodama, K. P., Jiang, W., Han, Y. L., Zhang, H., et al. (2010). New paleomagnetic results from the Lhasa Terrane: Revised estimation of latitudinal shortening across Tibet and implications for dating the India–Asia collision. *Earth and Planetary Science Letters*, *293*(3–4), 396–404. https://doi.org/10.1016/j.epsl.2010.03.013

Tapponnier, P., Xu, Z., Roger, F., Meyer, B., Arnaud, N., Wittlinger, G., & Yang, J. (2001). Oblique Stepwise rise and growth of the Tibet Plateau. *Science*, *294*(5547), 1671–1677. https://doi.org/10.1126/science.105978

Tauxe, L. (2005). Inclination flattening and the geocentric axial dipole hypothesis. *Earth and Planetary Science Letters*, *233*(3–4), 247–261. https://doi.org/10.1016/j.epsl.2005.01.027

Tauxe, L., & Kent, D. V. (2004). A simplified statistical model for the geomagnetic field and the detection of shallow bias in paleomagnetic inclinations: Was the ancient magnetic field dipolar? In *Timescales of the paleomagnetic field* (Vol. 145, pp. 101–115). American Geophysical Union. https://doi.org/10.1029/145gm08

Tong, Y. B., & Pei, J. L. (2024). Thermal demagnetization and rock magnetic results for the Jurassic limestone of the Gyangze [Dataset]. *Figshare*. https://doi.org/10.6084/m9.figshare.25051322.v1

Tong, Y. B., Yang, Z. Y., Li, J. F., Pei, J. L., & Li, J. F. (2019). New insights into the collision process of India and Eurasia: Evidence from the syntectonic-sedimentation-induced inclinational divergence of Cretaceous paleomagnetic data of the Lhasa Terrane. *Earth-Science Reviews*, *190*, 570–588. https://doi.org/10.1016/j.earscirev.2019.02.009

Tong, Y. B., Yang, Z. Y., Pei, J. L., Li, J. F., Jin, S. C., Hou, L. F., et al. (2022). New Late Cretaceous paleomagnetic results from the Lhasa Terrane and its implications for the process of suturing of India and Eurasia and close of the Neo-Tethys Ocean. *Geological Society of America Bulletin*, *134*(11/12), 3242–3257. https://doi.org/10.1130/B36310.1

Tong, Y. B., Yang, Z. Y., Zheng, L. D., Yang, T. S., Shi, L. F., Sun, Z. M., & Pei, J. L. (2008). Early Paleocene paleomagnetic results from southern Tibet, and tectonic implications. *International Geology Review*, *50*(6), 546–562. https://doi.org/10.2747/0020-6814.50.6.546

van Hinsbergen, D. J. J., Lippert, P. C., Dupont-Nivet, G., McQuarrie, N., Doubrovine, P. V., Spakman, W., & Torsvik, T. H. (2012). Greater India Basin hypothesis and a two-stage Cenozoic collision between India and Asia. *Proceedings of the National Academy of Sciences*, *109*(20), 7659–7664. https://doi.org/10.1073/pnas.1117262109

Wang, J. G., Hu, X. M., Garzanti, E., An, W., & Liu, X. C. (2017). The birth of the Xigaze forearc basin in southern Tibet. *Earth and Planetary Science Letters*, *465*, 38–47. https://doi.org/10.1016/j.epsl.2017.02.036

Weaver, R., Roberts, A. P., & Barker, A. J. (2002). A late diagenetic (syn-folding) magnetization carried by pyrrhotite: Implications for paleomagnetic studies from magnetic iron sulphide-bearing sediments. *Earth and Planetary Science Letters*, *200*(3–4), 371–386. https://doi.org/10.1016/S0012-821X(02)00652-0

Wu, F. L., Fang, X. M., Yang, Y. B., Dupont-Nivet, G., Nie, J. S., Fluteau, F., et al. (2022). Reorganization of Asian climate in relation to Tibetan Plateau uplift. *Nature Review Earth & Environment*, *3*(10), 684–700. https://doi.org/10.1038/s43017-022-00331-7

Wu, F. Y., Ji, W. Q., Wang, J. G., Liu, C. Z., Chung, S. L., & Clift, P. D. (2014). Zircon U-Pb and Hf isotopic constraints on the onset time of India-Asia collision. *American Journal of Science*, *314*(2), 548–579. https://doi.org/10.2475/02.2014.04

Xu, Y. C., Tan, X. D., Li, S., Li, Y. L., Ran, B., Han, Y. L., et al. (2022). Paleomagnetism of the Greater Indian passive margin sediments from the Upper Cretaceous succession: Evidence for long-delayed remagnetizations and implication for the India-Asia collision. *Journal of Asian Earth Sciences*, *229*, 105165. https://doi.org/10.1016/j.jseaes.2022.105165

Yang, T. S., Jin, J. J., Bian, W. W., Ma, Y. M., Gao, F., Peng, W. X., et al. (2019). Precollisional latitude of the Northern Tethyan Himalaya from the Paleocene Redbeds and its implication for Greater India and the India-Asia collision. *Journal of Geophysical Research: Solid Earth*, *124*(11), 10777–10798. https://doi.org/10.1029/2019JB017927

Yang, T. S., Ma, Y. M., Bian, W. W., Jin, J. J., Zhang, S. H., Wu, H. C., et al. (2015). Paleomagnetic results from the Early Cretaceous Lakang Formation lavas: Constraints on the paleolatitude of the Tethyan Himalaya and the India-Asia collision. *Earth and Planetary Science Letters*, *428*, 120–133. https://doi.org/10.1016/j.epsl.2015.07.040

Yi, Z., Huang, B., Chen, J., Chen, L., & Wang, H. (2011). Paleomagnetism of early Paleogene marine sediments in southern Tibet, China: Implications to onset of the India-Asia collision and size of Greater India. *Earth and Planetary Science Letters*, *309*, 153–165. https://doi.org/10.1016/j.epsl.2011.07.001

Yi, Z., Huang, B., Yang, L., Tang, X., Yan, Y., Qiao, Q., et al. (2015). A quasi-linear structure of the southern margin of Eurasia prior to the India-Asia collision: First paleomagnetic constraints from Upper Cretaceous volcanic rocks near the western syntaxis of Tibetan. *Tectonics*, *34*(7), 1431–1451. https://doi.org/10.1002/2014TC003571

Yin, A., & Harrison, T. M. (2000). Geologic evolution of the Himalayan-Tibetan orogen. *Annual Review of Earth and Planetary Sciences*, *28*(1), 211–280. https://doi.org/10.1146/annurev.earth.28.1.211

Yuan, J., Deng, C. L., Yang, Z. Y., Krijgsman, W., Thubtantsering, Qin, H. F., et al. (2022). Triple-stage India-Asia collision involving arc-continent collision and subsequent two-stage continent-continent collision. *Global and Planetary Change*, *212*, 103821. https://doi.org/10.1016/j.gloplacha.2022.103821

Yuan, J., Yang, Z. Y., Deng, C. L., Krijgsman, W., Hu, X. M., Li, S. H., et al. (2021). Rapid drift of the Tethyan Himalaya terrane before two-stage India-Asia collision. *National Science Review*, *8*, nwaa173. https://doi.org/10.1093/nsr/nwaa173

Zhao, Q., Huang, B. C., Yi, Z. Y., & Xue, P. F. (2021). High-resolution petrographic evidence confirming detrital and biogenic magnetites as remanence carriers for Zongpu Carbonates in the Gamba Area, South Tibet. *Frontiers in Earth Science*, *9*, 713469. https://doi.org/10.33389/feart.2021.713469

19448007, 2024, 15, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024GL110286 by Curtin University Library, Wiley Online Library on [13/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

References From the Supporting Information

Alken, P., Thébault, E., Beggan, C. D., Amit, H., Aubert, J., Baerenzung, J., et al. (2021). International geomagnetic reference field: The thirteenth generation. *Earth Planets and Space*, *73*(1), 49. https://doi.org/10.1186/s40623-020-01288-x

Dekkers, M. J., Mattei, J. L., Fillion, G., & Rochette, P. (1989). Grain-size dependence of the magnetic behavior of pyrrhotite during its low temperature transition at 34 K. *Geophysical Research Letters*, *16*(8), 855–858. https://doi.org/10.1029/GL016i008p00855

Fisher, R. A. (1953). Dispersion on a sphere. *Proceedings of the Royal Society of London, Series A*, *217*(1130), 295–305. https://doi.org/10.1098/rspa.1953.0064

Harrison, R. J., & Feinberg, J. M. (2008). FORCinel: An improved algorithm for calculating first-order reversal curve distributions using locallyweighted regression smoothing. *Geochemistry, Geophysics, Geosystems*, *9*(5), Q05016. https://doi.org/10.1029/2008GC001987

Kirschvink, J. L. (1980). The least-squares line and plane and the analysis of paleomagnetic data. *Geophysical Journal of the Royal Astronomical Society*, *62*(3), 699–718. https://doi.org/10.1111/j.1365-246X.1980.tb02601.x

Kosterov, A. (2003). Low-temperature magnetization and AC susceptibility of magnetite: Effect of thermomagnetic history. *Geophysical Journal International*, *154*(1), 58–71. https://doi.org/10.1046/j.1365-246X.2003.01938.x

Lowrie, W. (1990). Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties. *Geophysical Research Letters*, *17*(2), 159–162. https://doi.org/10.1029/GL017i002p00159

McFadden, P. L. (1990). A new fold test for paleomagnetic studies. *Geophysical Journal International*, *103*(1), 163–169. https://doi.org/10.1111/j.1365-246X.1990.tb01761.x

Roberts, A. P. (1995). Magnetic properties of sedimentary greigite (Fe_3S_4). *Earth and Planetary Science Letters*, *134*(3–4), 227–236. https://doi.org/10.1016/0012-821X(95)00131-U

Roberts, A. P., Liu, Q. S., Rowan, C. J., Chang, L., Carvallo, C., Torrent, J., & Horng, C. S. (2006). Characterization of hematite (a-Fe_2O_3), goethite (a-FeOOH), greigite (Fe_3S_4), and pyrrhotite (Fe_7S_8) using first-order reversal curve diagrams. *Journal of Geophysical Research*, *111*(B12), B12S35. https://doi.org/10.1029/2006JB004715

Rochette, P., Fillion, G., Mattéi, J. L., & Dekkers, M. J. (1990). Magnetic transition at 30-34 kelvin in pyrrhotite: Insight into a widespread occurrence of this mineral in rock. *Earth and Planetary Science Letters*, *98*(3–4), 319–328. https://doi.org/10.1016/0012-821X(90)90034-U

Watson, G. S., & Enkin, R. J. (1993). The fold test in paleomagnetism as a parameter estimation problem. *Geophysical Research Letters*, *20*(19), 2135–2137. https://doi.org/10.1029/93GL01901

Zijderveld, J. D. A. (1967). A.C. demagnetization of rocks: Analysis of results. In D. W. Collinson, K. M. Creer, & S. K. Runcorn (Eds.), *Methods in paleomagnetism* (pp. 254–286). Elsevier Pres.

19448007, 2024, 15, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024GL110286 by Curtin University Library, Wiley Online Library on [13/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License